

Continuous Distributions

Probability is area under a curve

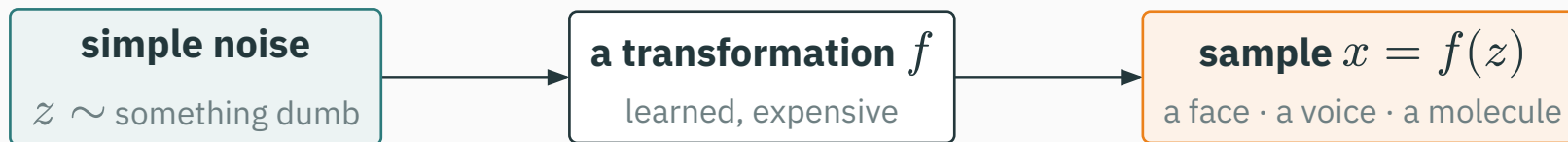
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Mathematical Foundations for AI · IIT Gandhinagar

The hook: machines that draw from noise

Where does a brand-new face come from?

Image generators, voice cloners, molecule designers — none of them stores a warehouse of faces to hand out. Every fresh sample is manufactured on the spot, by one recipe:



That first box is astonishingly humble. In PyTorch it is literally

```
z = torch.randn(...)    # plain Gaussian noise — that's the whole "creativity"  
x = f(z)                # a learned transformation does the rest
```

Diffusion models, VAEs, normalizing flows — different learned f , **same recipe**: sample cheap noise, transform it.

Two questions hide inside that recipe

1. The noise z and the sample x are **continuous** quantities. In ES 114, probability meant counting outcomes: $P(X = x)$. What does probability even **mean** when x can be any real number?
2. The transformation f moves samples around. What does it do to their **probabilities**?

Honesty up front: the real f is a neural network in a million dimensions (ES 667's story). Today we build the **1-D mathematics that makes the recipe legal** — and by the last section you will run the recipe yourself, exactly, on a real distribution.

Today, in one line

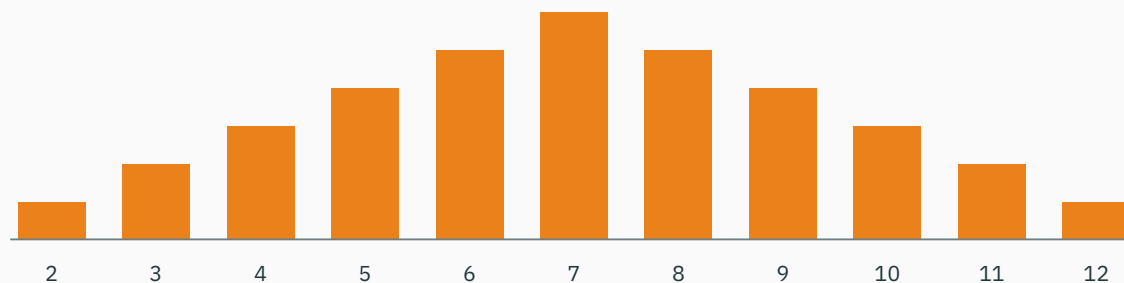
For continuous quantities, probability lives in areas under a density curve — and any transformation of the variable must pay a Jacobian toll.

The route:

1. **Density** — the object that replaces the PMF (and the traps it sets)
2. **CDF** — the running total that connects density to real probabilities
3. **Expectation & variance** — the old formulas, with $\sum \rightarrow \int$
4. **The zoo** — uniform, exponential, Gaussian, Beta
5. **The toll** — change of variables, then **sampling anything from** $\mathcal{U}(0, 1)$

What ES 114 already gave you

Two dice, Z = sum of the faces — the PMF you have drawn many times:



Everything lived on this scaffolding: $p_Z(z) = P(Z = z)$ is a genuine probability, probabilities of events are **sums** of bar heights, $\mathbb{E}[Z] = \sum z p_Z(z)$.

Today every one of those moves gets a continuous twin — and two of them change in ways that surprise almost everyone.

Learning outcomes

By the end of this lecture you will be able to:

1. Explain why $P(X = x) = 0$ for continuous X — and why a **density** fixes it.
2. Read a pdf correctly: probability = **area**; densities may exceed 1.
3. Convert between **pdf and CDF** in both directions (integrate / differentiate).
4. Compute **expectations and variances** as integrals (uniform, exponential).
5. Recognize the zoo on sight: **uniform, exponential, Gaussian, Beta**.
6. Derive the 1-D **change-of-variables** rule $p_Y(y) = p_X(x)|dx/dy|$.
7. Prove **inverse-CDF sampling** correct and implement it in 5 lines of NumPy.

**Density: probability per unit
length**

A question ES 114 cannot answer

ES 114's dataset of **25,000 measured heights**. Pick a person at random; let H be their height.

Easy question: $P(66 \leq H \leq 70)$? Count the rows, divide by 25,000. Fine.

Treacherous question: what is

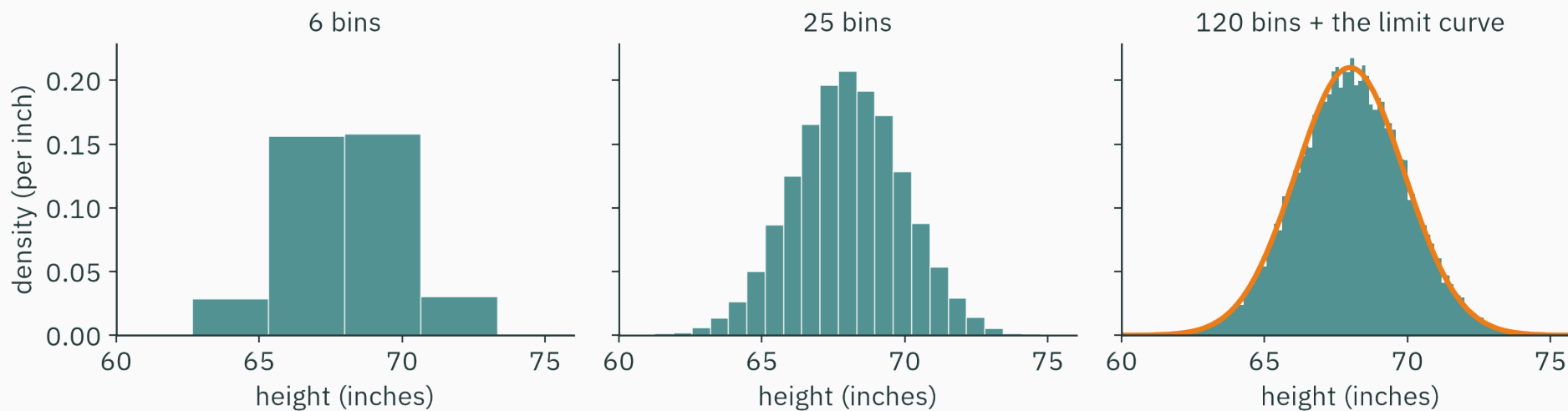
$$P(H = \textit{exactly} 68 \text{ inches}) = P(H = 68.000000000...) = ?$$

Nobody is **exactly** 68 inches. Any specific real number has infinitely many decimals to match; matching all of them is a miracle of probability zero.

For a continuous quantity, $P(X = x) = 0$ at **every single** x . The PMF — our whole ES 114 toolkit — reads 0 everywhere. We need a new object.

Picture: squeeze the histogram until it becomes a curve

Same 25,000 heights, plotted as **density histograms** (bar area = fraction of data), with ever finer bins:



The staircase stops changing shape and settles onto a smooth curve. **That limiting curve** — not any single bar — is the new object.

The limit object: a probability density

$$p(x) = \lim_{\delta \rightarrow 0} \frac{P(x \leq X \leq x + \delta)}{\delta}$$

— probability **per unit length**, not probability.

Read it like the speedometer of probability: for a tiny interval,

$$P(x \leq X \leq x + \delta) \approx p(x) \cdot \delta \quad (\text{height} \times \text{width} = \text{a sliver of area})$$

- The units are real: our height curve peaks at $p \approx 0.21$ **per inch**.
- Probability of a **point**: width $\delta \rightarrow 0$, so $P(X = x) = 0$ — the paradox is now a feature, not a bug.

Zero probability is not “impossible”

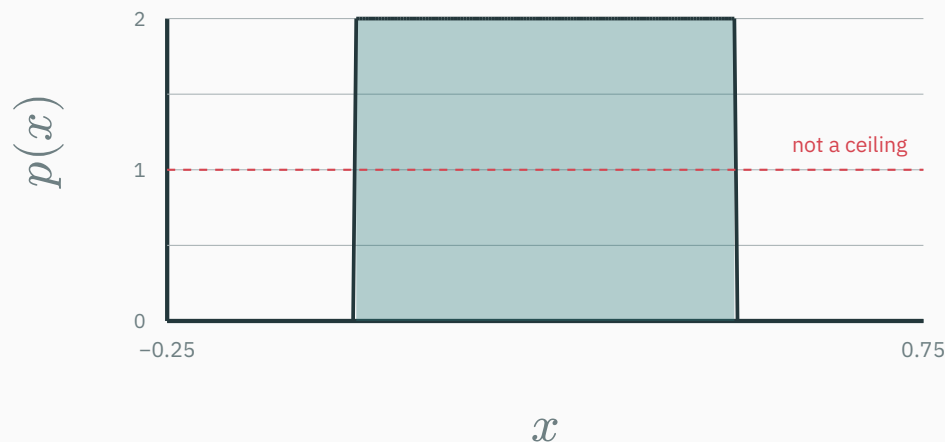
Uncomfortable but true: the dart **does** land somewhere, yet every exact landing point had probability 0.

- “Probability 0” for continuous X means **vanishingly unlikely under infinite precision** — not forbidden. Some zero-probability outcome occurs on every draw.
- Nothing is broken: probability was never promised to individual points, only to **sets with width**.

This is the door to measure theory — which we will keep respectfully closed. MML §6.1 does the same; the pictures-and-areas view is all this course (and most of ML) needs.

The first trap: a density of 2

$X \sim \mathcal{U}(0, 1/2)$ — all mass spread evenly on an interval of width $1/2$:

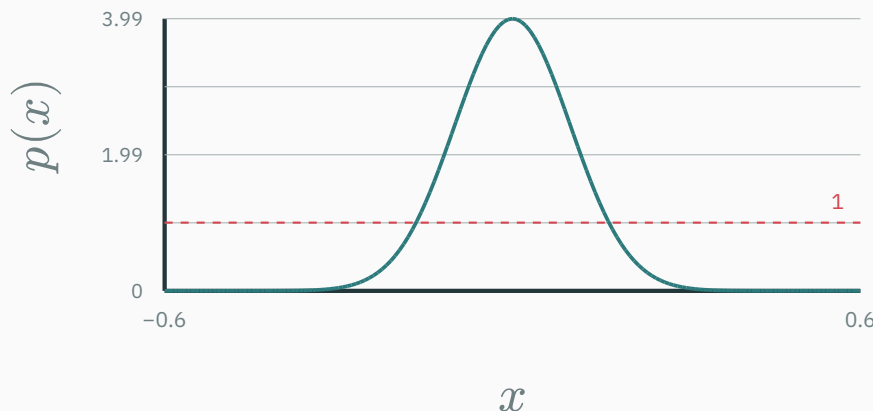


The height is $p(x) = 2$ — **bigger than 1** — yet nothing is wrong: $\text{area} = 2 \times 1/2 = 1$. ✓

Densities are not probabilities. $p(x) > 1$ is legal, common, and means only: mass is concentrated.

The trap, sprung twice

A second offender: $\mathcal{N}(0, \sigma = 0.1)$, a Gaussian squeezed tight —



Peak value $1/(0.1\sqrt{2\pi}) \approx 3.99$. Shrink σ and the peak grows **without bound** — the area stays exactly 1.

Spot-the-bug skill for later courses: a “probability” of 2.7 in your log output is **fine** if it is a density (`log_prob` of a narrow Gaussian, say) — and a catastrophe if it is supposed to be a probability. Know which one you are holding.

Numbers: three questions on $\mathcal{U}(0, 1/2)$

Height 2 on $[0, 1/2]$, zero elsewhere. Probability = area = height $\times$ width:

Question	Area computation	Answer
$P(0.2 \leq X \leq 0.3)$	2×0.1	0.2
$P(X = 0.25)$	2×0	0
$P(0 \leq X \leq 1)$	$2 \times 0.5 + 0 \times 0.5$	1

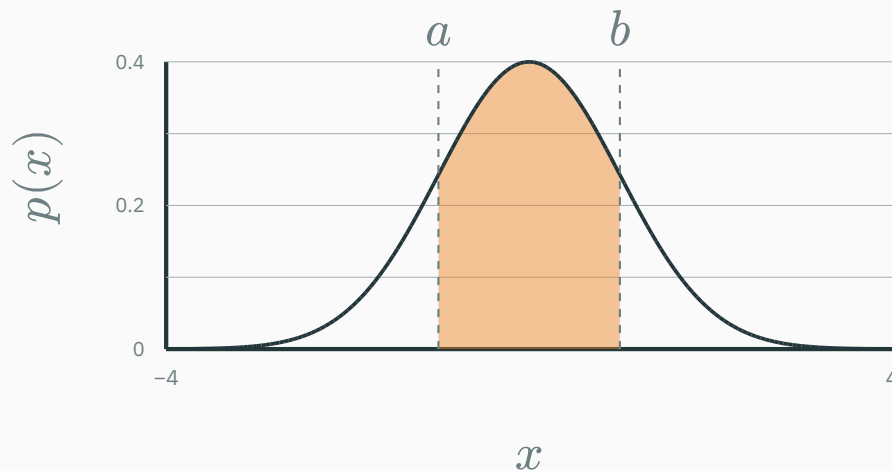
Row 2 in words: the tallest density in the world still assigns zero to a **point** — width is everything.

Row 3 in words: integrating past the support costs nothing — the density is 0 out there.

Probability is area, officially

The ES 114 axioms, translated for a density p :

$$p(x) \geq 0 \quad \text{everywhere,} \quad \int_{-\infty}^{\infty} p(x) \, dx = 1, \quad P(a \leq X \leq b) = \int_a^b p(x) \, dx$$



Nonnegative curve, total area 1, questions answered by shading. That's the whole contract.

Code: a density is just a function you can call

```
from scipy.stats import uniform, norm

u = uniform(loc=0, scale=0.5)          # U(0, 1/2)
u.pdf(0.25)                            # 2.0  ← density above 1; no alarm rings
u.cdf(0.3) - u.cdf(0.2)                # 0.2  ← an actual probability

norm(0, 0.1).pdf(0.0)                  # 3.989 — the squeezed Gaussian's peak
```

- .pdf returns **heights** — compare, plot, take logs, but never call them probabilities.
- Real probabilities come from .cdf differences — areas, exactly as the axioms demand. (The CDF is next.)

$X \sim \mathcal{U}(0, 1/2)$, **so** $p(x) = 2$ **on** $[0, 1/2]$. Which statement is **true**?

A. $P(X = 1/4) = 2$ — the density at that point

B. $p(1/4) = 2$ **and** $P(X = 1/4) = 0$ — no contradiction

C. A valid pdf can never exceed 1, so $\mathcal{U}(0, 1/2)$ is not a real distribution

D. $P(0 \leq X \leq 1/2) = 2$, because height 2 covers the whole support

Correct: B — $p(1/4) = 2$ and $P(X = 1/4) = 0$ coexist happily

Why: The density is probability **per unit length**: multiply by a width to get probability. A point has width 0, so $P(X = 1/4) = 2 \times 0 = 0$, while the height happily sits at 2. (A is heights-as-probabilities, C is the ceiling myth, and D would make total probability 2 — areas, never heights.)

The CDF: probability's running total

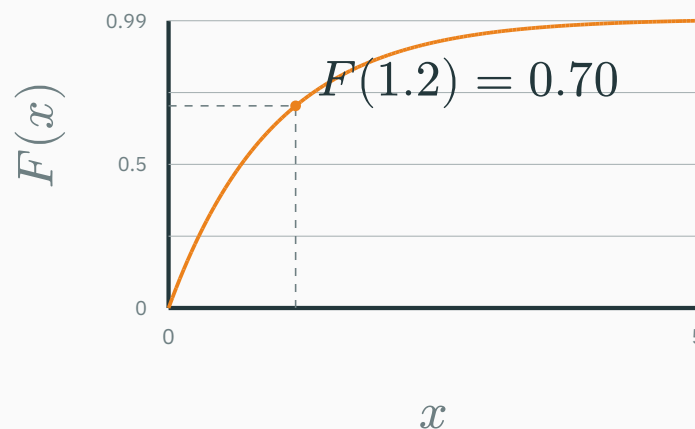
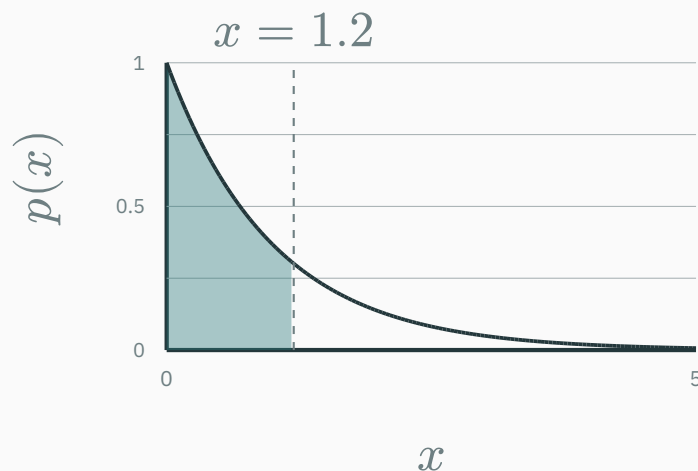
One function sidesteps the whole paradox

$F(x) = P(X \leq x)$ — the *cumulative distribution function*

- $F(x)$ is a **genuine probability** for every x — the event $X \leq x$ has width, no paradox.
- It climbs from 0 (far left) to 1 (far right), never decreasing — a running total of mass.
- And it is **one definition for every random variable** — discrete, continuous, or mixed. The PMF/pdf split disappears at the CDF level.

ES 114 flashback: you met CDFs as staircases — flat, then a jump of height $p_Z(z)$ at each outcome. Today's CDFs are smooth ramps; the jumps have melted.

Exp(1): density $p(x) = e^{-x}$, running total $F(x) = 1 - e^{-x}$.



The shaded area on the left **is** the marked height on the right. Same information, two displays: p shows **where** mass sits; F shows **how much** has accumulated.

Direction 1: integrate the density $\rightarrow$ CDF

The running total is the area collected so far:

$$F(x) = \int_{-\infty}^x p(t) dt$$

Worked on $\text{Exp}(1)$, every step visible:

$$F(x) = \int_0^x e^{-t} dt = [-e^{-t}]_0^x = (-e^{-x}) - (-e^0) = 1 - e^{-x}$$

Numbers: $F(1.2) = 1 - e^{-1.2} = 1 - 0.301 = 0.699$ — the 0.70 read off the picture. $\checkmark$

Direction 2: differentiate the CDF $\rightarrow$ density

Module 2's fundamental theorem, cashing a cameo: the derivative of accumulated area is the height being added —

$$F'(x) = p(x)$$

Round trip on $\text{Exp}(1)$:

$$\frac{d}{dx}(1 - e^{-x}) = e^{-x} = p(x) \quad \checkmark$$

pdf $\xrightarrow{\int}$ CDF and CDF $\xrightarrow{d/dx}$ pdf. One object, two views, both directions open.

This is also the honest **definition** of the density: whatever function you must integrate to reproduce F .

Every interval question routes through F

$$P(a < X \leq b) = F(b) - F(a) \quad (\text{total so far at } b, \text{ minus total so far at } a)$$

Worked on $\text{Exp}(1)$ — the chance the wait lands between 1 and 2:

$$F(2) - F(1) = (1 - e^{-2}) - (1 - e^{-1}) = e^{-1} - e^{-2} = 0.368 - 0.135 = 0.233$$

No new integral was computed — the CDF pre-paid it. This is why libraries ship `.cdf` and let you subtract.

Whether you write $<$ or $\leq$ changes nothing here: the endpoints carry probability 0. (For discrete variables it **does** matter — the staircase jumps.)

Code: .cdf answers real questions

Marks modeled as $\mathcal{N}(70, 8^2)$ — the toy model from ES 114's notebook:

```
from scipy.stats import norm
marks = norm(70, 8)

marks.cdf(60)                # 0.1056 → ~11% score below 60
marks.cdf(80) - marks.cdf(60) # 0.7887 → ~79% land in [60, 80]
1 - marks.cdf(90)           # 0.0062 → the A+ tail
```

Three managerial questions, three CDF calls, zero integrals by hand — the Gaussian's CDF has no closed formula, so **everyone** routes through the library.

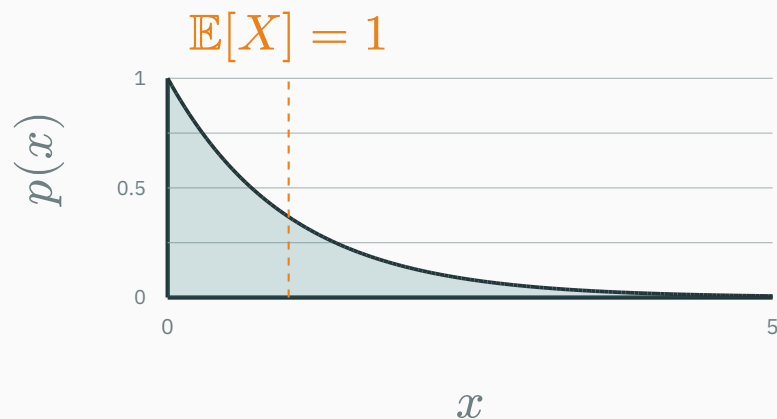
Expectation & variance, as integrals

The dictionary move: $\sum$ becomes $\int$

ES 114: $\mathbb{E}[X] = \sum_x x p_X(x)$. Continuous twin: chances become slivers $p(x) dx$ —

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x p(x) dx \qquad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) p(x) dx$$

Same picture as ever: $\mathbb{E}[X]$ is the **balance point** of the density cutout:



Trap dodged: the balance point is **not** the peak (mode, at 0) — the long right tail drags it out to 1.

Worked: the uniform's balance point

$X \sim \mathcal{U}(a, b)$, so $p(x) = 1/(b - a)$ on $[a, b]$:

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b - a} dx = \frac{1}{b - a} \cdot \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b - a)}$$

Factor the difference of squares:

$$\mathbb{E}[X] = \frac{(b - a)(b + a)}{2(b - a)} = \frac{a + b}{2}$$

The midpoint — exactly what symmetry promised. The integral just confirms the picture.

For $\mathcal{U}(0, 1)$: $\mathbb{E}[X] = 1/2$.

Worked: the exponential waits $1/\lambda$

$X \sim \text{Exp}(\lambda)$, $p(x) = \lambda e^{-\lambda x}$ on $[0, \infty)$. Integrate by parts ($u = x$, $dv = \lambda e^{-\lambda x} dx$):

$$\mathbb{E}[X] = \int_0^\infty x \lambda e^{-\lambda x} dx = \underbrace{[-x e^{-\lambda x}]_0^\infty}_{=0} + \int_0^\infty e^{-\lambda x} dx$$

$$\mathbb{E}[X] = \left[-\left(\frac{1}{\lambda}\right) e^{-\lambda x} \right]_0^\infty = \frac{1}{\lambda}$$

Sanity: λ is a **rate** (events per unit time), so the mean wait is $1/\lambda$ time units — buses arriving at rate 2 per hour are a half-hour wait on average. Units cooperate; always check they do.

Variance is an integral too

Spread = expected squared distance from the balance point:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \int (x - \mu)^2 p(x) \, dx = \mathbb{E}[X^2] - \mu^2$$

Worked on $\mathcal{U}(0, 1)$ via the shortcut:

$$\mathbb{E}[X^2] = \int_0^1 x^2 \, dx = \frac{1}{3}, \quad \text{Var}(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12} \approx 0.083$$

And the exponential: $\text{Var}(X) = 1/\lambda^2$ (same integration-by-parts trick twice — practice problem 3).

Old friends survive unchanged: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ and $\text{Var}(aX + b) = a^2 \text{Var}(X)$ — linearity never cared whether the sum was $\sum$ or $\int$.

Code: Monte Carlo audits our integrals

Can't integrate? **Sample and average** — the mean of draws approximates $\mathbb{E}[X]$:

```
rng = np.random.default_rng(0)

x = rng.uniform(0, 1, 1_000_000)
x.mean(), x.var()           # (0.50002, 0.08333)   = 1/2, 1/12   ✓

t = rng.exponential(1.0, 1_000_000)
t.mean(), t.var()           # (0.99950, 0.99852)   = 1/λ, 1/λ²   ✓
```

Both hand-derived integrals confirmed to three digits by brute randomness.

Sampling is an integral evaluator. Half of modern ML lives on this one sentence — and it needs a **sampler, which the last section builds.**

The zoo: four cards to know on sight

How to read a zoo card

Choosing a distribution = choosing **where mass is allowed to live** (the support), then **how it is shaped** (the knobs):

Name	Support	Knobs	Typical job in ML
Uniform $\mathcal{U}(a, b)$	$[a, b]$	endpoints	raw randomness; initializations
Exponential $\text{Exp}(\lambda)$	$[0, \infty)$	rate λ	waiting times, positive scales
Gaussian $\mathcal{N}(\mu, \sigma^2)$	$(-\infty, \infty)$	center, spread	noise, priors, embeddings — everything
Beta(α, β)	$[0, 1]$	two shapes	beliefs about a probability

Support first, always: a Gaussian for waiting times leaks mass into negative time; a Beta can't model heights. Match the room before decorating it.

Card 1 · Uniform: the raw material

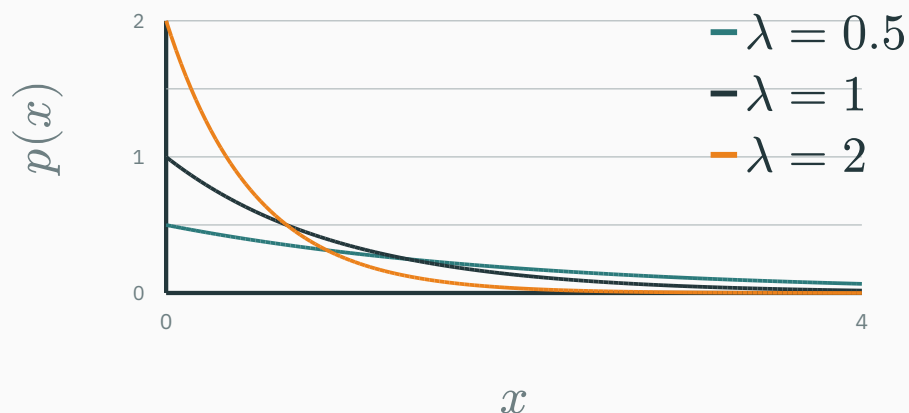
$$X \sim \mathcal{U}(a, b) \quad p(x) = \begin{cases} 1/(b-a) & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \quad \mathbb{E}[X] = \frac{a+b}{2}, \quad \text{Var} = \frac{(b-a)^2}{12}$$

- The CDF is a straight ramp: $F(x) = (x-a)/(b-a)$ on the support.
- Fair dice made continuous: no value in the support is more special than another.

$\mathcal{U}(0, 1)$ **is what `rng.random()` gives you — and by the end of today it is the *only* randomness you will ever need.**

Card 2 · Exponential: waiting times

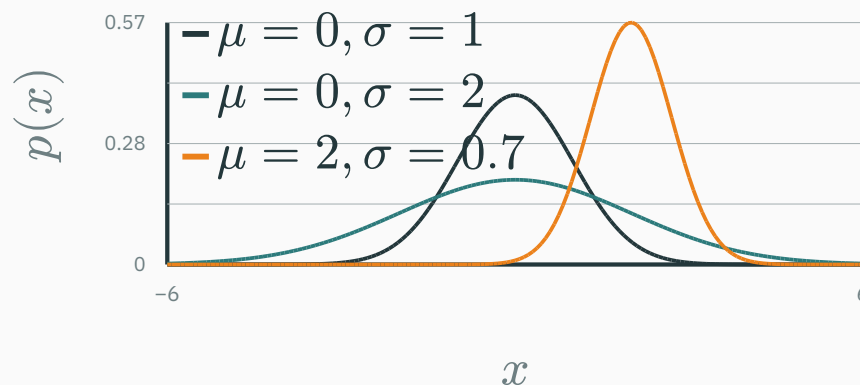
$$X \sim \text{Exp}(\lambda) \quad p(x) = \lambda e^{-\lambda x} \quad \text{on } [0, \infty) \quad \mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var} = \frac{1}{\lambda^2}$$



Time between requests at a server, between clicks, between mutations. One famous quirk: it is **memoryless** — having already waited 5 minutes changes nothing about the wait ahead.

Card 3 · Gaussian: the celebrity

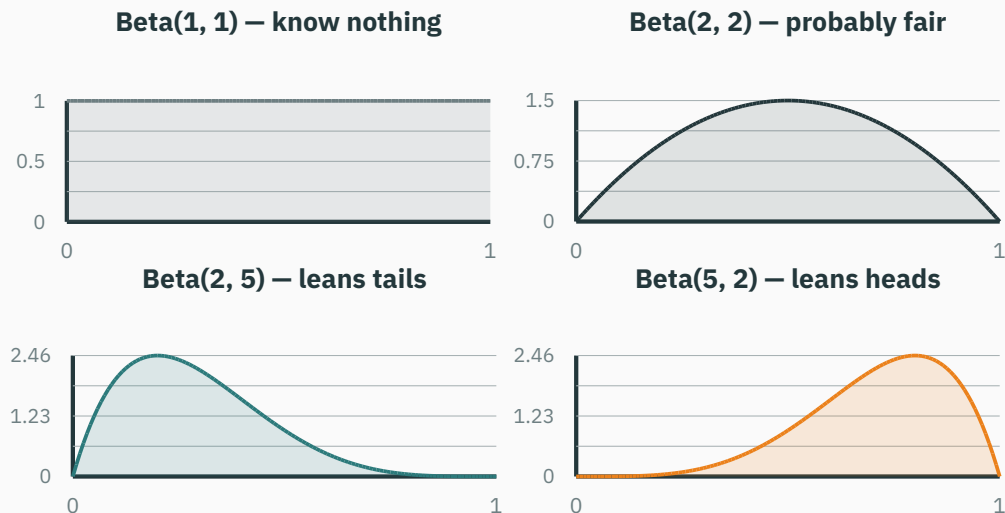
$$X \sim \mathcal{N}(\mu, \sigma^2) \quad p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$



μ slides, σ stretches — that is the **entire** family. Tape measure: $\approx 68\%$ of mass within 1σ , 95% within 2σ , 99.7% within 3σ .

Why everywhere? Sums of many small effects become Gaussian (CLT — demoed in L13), and it is the “most honest” default (max entropy — proved in the information module).

Beta draws live in $[0, 1]$ — so a draw can **be** a probability (a coin's heads-rate θ):



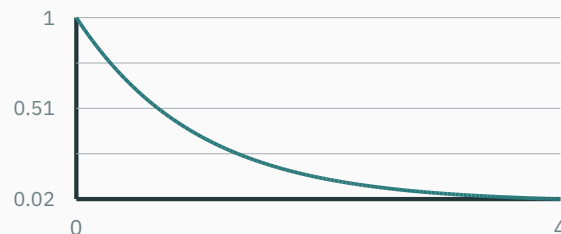
$p(\theta) \propto \theta^{\alpha-1} (1-\theta)^{\beta-1}$, with $\mathbb{E}[\theta] = \alpha / (\alpha + \beta)$. **Planted for L15:** read $\alpha - 1$ as “heads seen”, $\beta - 1$ as “tails seen” — this becomes the machinery of Bayesian updating, with the Beta as star. Remember the shapes.

The four, on one wall

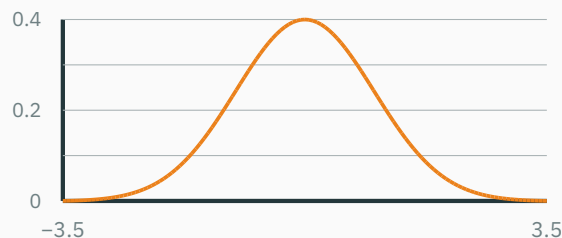
Uniform — flat



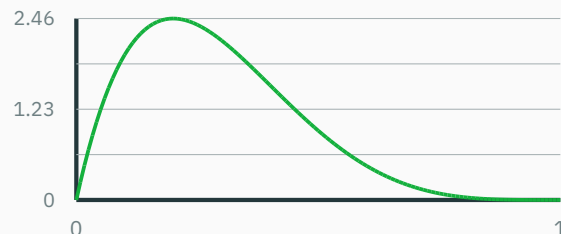
Exponential — decaying



Gaussian — the bell



Beta — shapes on $[0, 1]$



Flat box, falling slide, bell, and the shape-shifter on $[0, 1]$. From these four (plus L13's multivariate bell) you can read 90% of the distributions in ML papers.

INTERACTIVE

↗ distribution-explorer.github.io/

Distribution Explorer — every distribution in tonight’s reading with sliders for each knob. Watch pdf and CDF move together; find the parameter settings that push a density above 1.

INTERACTIVE

↗ seeing-theory.brown.edu/

Seeing Theory (Brown) — the “Probability Distributions” and “Random Variables” chapters animate exactly today’s arc: PMF → pdf, area = probability, CDF as running total. 15 beautiful minutes.

The Jacobian toll: change of variables

The hook returns: transform the noise

The generative recipe was: $z \sim$ simple noise, $x = f(z)$. For the recipe to be **mathematics** rather than vibes, we must answer:

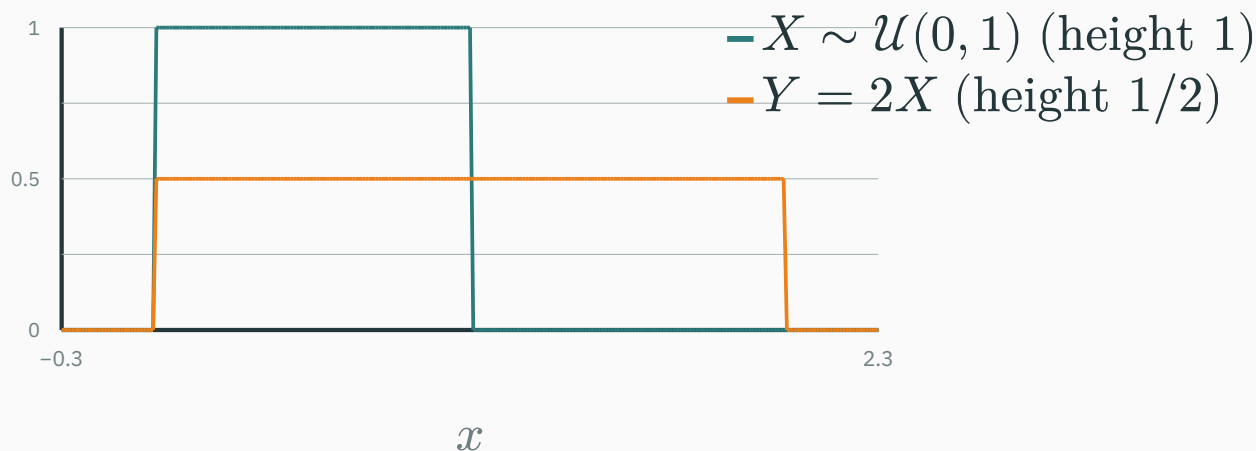
If X has density p_X and $Y = g(X)$ — what is p_Y ?

Start with the humblest transformation and be **completely** sure of it:

$$Y = aX + b \quad (\text{stretch by } a, \text{ slide by } b)$$

Concrete pilot case: $X \sim \mathcal{U}(0, 1)$ and $Y = 2X$. Before any formula — what **should** p_Y look like?

Picture: stretch the axis, squash the curve



Y lives on $[0, 2]$ — twice the room. Total probability is still 1 — same area. Twice the width forces **half the height**.

Stretching the axis by a dilutes the density by a . Area is conserved; height pays.

Numbers: follow one interval through the stretch

Track the fate of $[0.2, 0.3]$ under $y = 2x$:

- Its probability under X : width 0.1, height 1 $\rightarrow P = 0.1$.
- Where does it land? $[0.4, 0.6]$ — the **same** outcomes, relabeled. The 0.1 of probability rides along untouched.
- New width 0.2. For the area to still be 0.1: height $= 0.1/0.2 = 1/2$. $\checkmark$ matches the picture.

Probability belongs to **events** (sets of outcomes), and a transformation only renames outcomes — so event probabilities are frozen. Densities are the ones that must adjust, because **widths** changed underneath them.

The tempting wrong answer

“Just substitute”: $p_Y(y) = p_X(y/2)$?

Test it. $p_X(y/2) = 1$ for all $y \in [0, 2]$ — a box of height 1 over width 2:

$$\int_0^2 1 \, dy = 2 \quad \text{— total probability 2. Not a distribution.}$$

Densities do **not** transform by substitution alone. Substitution relabels the x -axis but forgets that the axis itself was stretched — the area bookkeeping is missing. Forgetting this factor is the classic hand-rolled-likelihood bug.

The fix must multiply by exactly the factor that repairs the area. Let's derive it, not guess it.

★ The derivation, step 1: route through the CDF

Densities are treacherous; **CDFs are genuine probabilities** — safe to push around.

So compute F_Y first. Take $Y = aX + b$ with $a > 0$:

$$F_Y(y) = P(Y \leq y) = P(aX + b \leq y)$$

Solve the inequality for X (allowed: $a > 0$ keeps the direction):

$$F_Y(y) = P\left(X \leq \frac{y - b}{a}\right) = F_X\left(\frac{y - b}{a}\right)$$

No approximation, no density yet — just the event $\{Y \leq y\}$ renamed as an event about X . This “CDF first, then differentiate” move is **the** standard tool for transformed variables.

★ Step 2: differentiate — the chain rule cashes in

Density = derivative of the CDF (Direction 2, minutes old):

$$p_Y(y) = \frac{d}{dy} F_X\left(\frac{y-b}{a}\right) = p_X\left(\frac{y-b}{a}\right) \cdot \underbrace{\frac{1}{a}}_{\text{chain rule's inner derivative}}$$

If $a < 0$ the inequality flips ($P(X \geq \dots) = 1 - F_X$), and differentiating brings $-1/a$ — positive again. Both signs in one formula:

$$p_Y(y) = p_X\left(\frac{y-b}{a}\right) \cdot \frac{1}{|a|} \quad \text{— the stretch factor, paid in density}$$

★ Step 3: interrogate the formula

Check against the pilot: $X \sim \mathcal{U}(0, 1)$, $Y = 2X$: $p_Y(y) = p_X(y/2) \cdot 1/2 = 1/2$ on $[0, 2]$ — the half-height box we measured by hand. ✓

Check the area is repaired: substitute $x = (y - b)/a$, $dy = |a| dx$:

$$\int p_X\left(\frac{y-b}{a}\right) \frac{1}{|a|} dy = \int p_X(x) \frac{1}{|a|} |a| dx = \int p_X(x) dx = 1 \quad \checkmark$$

The $1/|a|$ is precisely what the substitution rule of integration (Module 2) demands — change of variables for densities **is** change of variables for integrals, wearing a probability costume.

The general 1-D rule

For any strictly monotone, differentiable g , the same two moves (CDF, then chain rule) give:

$$p_Y(y) = p_X(g^{-1}(y)) \cdot \left| \frac{dx}{dy} \right| \quad \text{where } x = g^{-1}(y)$$

Read it aloud: **old density at the source point, times how much the axis was locally stretched**. For linear g , the stretch $|dx/dy| = 1/|a|$ everywhere; for curved g it varies point by point.

Worked cameo you already met in ES 114's notebook: $X = e^Z$ with Z Gaussian gives the **log-normal**, whose density carries a $1/x$ — that $1/x$ is exactly $|dz/dx|$. What looked like a mysterious extra factor is the toll.

L9's toll booth: you are the customer

L9 ended with a promise — “densities that change variables must pay the $|\det J|$ area toll. Same dial, new customer.”

L9 (geometry)	L12 (probability)
map $\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$	transform $g : \mathbb{R} \rightarrow \mathbb{R}$
little square $\rightarrow$ parallelogram	little interval $\rightarrow$ stretched interval
$ \det J $ = local area scaling	$ \mathrm{d}y / \mathrm{d}x $ = local length scaling
areas multiply by $ \det J $	densities divide by it: $p_Y = p_X \cdot \mathrm{d}x / \mathrm{d}y $

In d dimensions the rule is verbatim $p_Y(\mathbf{y}) = p_X(\mathbf{x})|\det J|^{-1}$ — the factor every normalizing-flow and diffusion paper carries in its log-likelihood. You have now derived its 1-D soul.

Payoff: build any Gaussian from the standard one

Let $Z \sim \mathcal{N}(0, 1)$ — density $\varphi(z) = e^{-z^2/2} / \sqrt{2\pi}$ — and stretch-and-slide: $X = \mu + \sigma Z$.
The toll rule with $a = \sigma$, $b = \mu$:

$$p_X(x) = \varphi\left(\frac{x - \mu}{\sigma}\right) \cdot \frac{1}{\sigma} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

That is the $\mathcal{N}(\mu, \sigma^2)$ density, **derived** rather than memorized — the $1/\sigma$ out front is a Jacobian toll and always was.

```
z = torch.randn(10_000)      # N(0, 1) — the only Gaussian anyone samples
x = mu + sigma * z           # N(mu, sigma^2) — transformation does the rest
```

This exact two-liner is the **reparameterization trick** inside every VAE — hook, meet mathematics.

$X \sim \mathcal{U}(0, 1)$ **and** $Y = 3X$. **What is** $p_Y(1.5)$?

A. 1 — uniform densities stay height 1

B. 3 — the stretch multiplies the density by 3

C. $1/3$ — the stretch dilutes the density by 3

D. 0 — $y = 1.5$ is outside the support of Y

Correct: C — $p_Y(1.5) = 1 \cdot 1/|3| = 1/3$

Why: Y lives on $[0, 3]$ (so D is out), and stretching by 3 spreads the same area over triple the width — density **divides** by 3 (so B has the toll backwards, and A forgot the toll entirely). Sanity: height $1/3 \times$ width $3 =$ area 1 . ✓

**Inverse-CDF sampling: uniform
→ anything**

The last missing piece: where do samples come from?

All of today assumed we can “draw $x \sim p$ ”. But a computer owns exactly **one** source of randomness:

```
u = rng.random()      # U(0, 1) – this is ALL the machine truly has
```

`randn`, `exponential`, every fancy sampler — all are **transformations of uniform draws**. There is no other well.

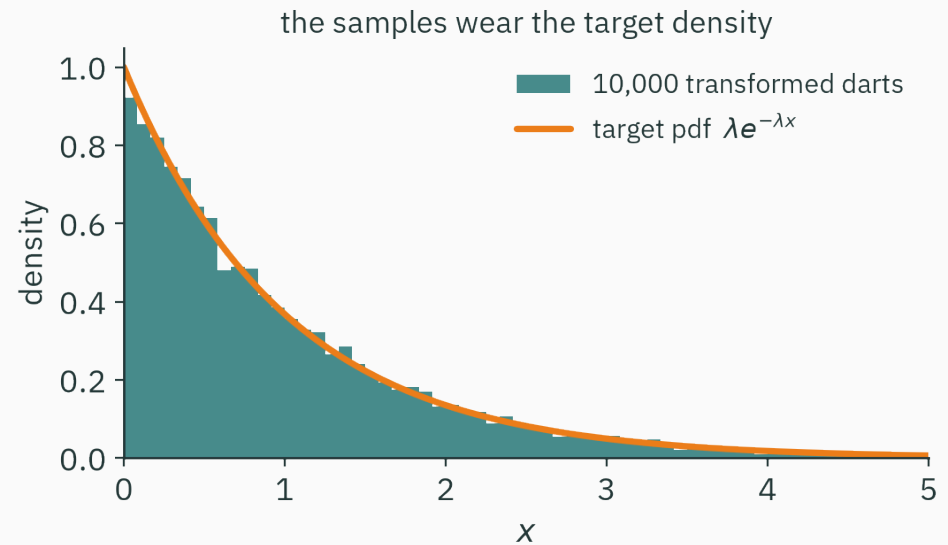
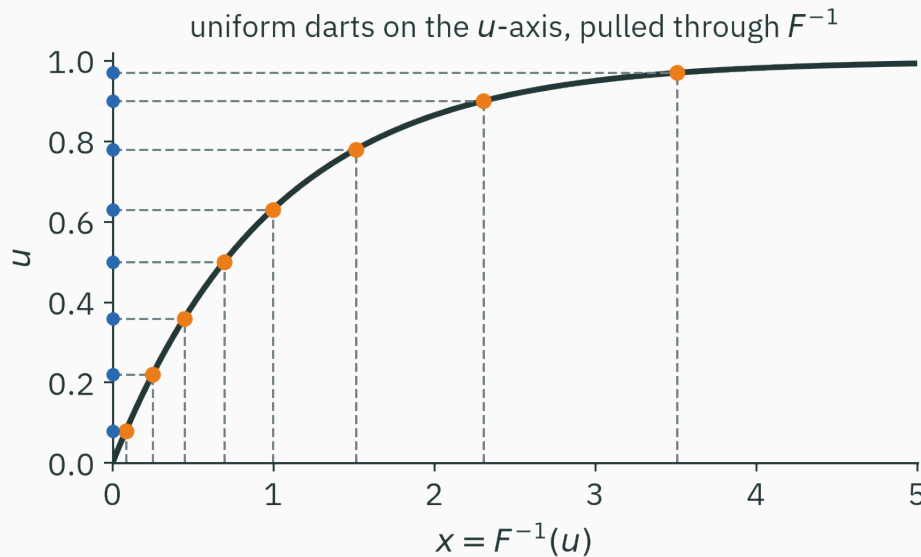
Claim: $\mathcal{U}(0, 1)$ + a CDF F = a sampler for F 's distribution. The transform is

$$x = F^{-1}(u).$$

One transformation turning dumb noise into a chosen distribution — the generative recipe, in its smallest honest form.

The picture: darts at the u -axis

Throw uniform darts at the **vertical** axis of the CDF; pull each through F^{-1} down to the x -axis:



Left: equal-density darts on $[0, 1]$. Right: their landing spots wear the exponential's density exactly.

Why it works, in words first

Where the CDF is **steep**, a wide band of u -values funnels into a narrow band of x — many darts, small region: **high density**.

Where the CDF is **flat**, a narrow band of u spreads over a huge range of x — few darts per unit length: **low density** (the exponential's tail).

Steepness of F is $F' = p$ — the density itself. The funnel automatically deals darts in proportion to p . That is the entire mechanism; now we make it a theorem.

★ Correctness, in two lines

D DERIVATION

Let $U \sim \mathcal{U}(0, 1)$ and define $X = F^{-1}(U)$, with F strictly increasing. Compute X 's CDF:

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) \quad (\text{apply the increasing map } F \text{ to both sides})$$

And for a uniform, $P(U \leq t) = t$ for any $t \in [0, 1]$:

$$P(X \leq x) = F(x) \quad \blacksquare$$

$F^{-1}(U)$ has CDF F — **exactly**, not approximately. Two lines, no fine print.

(Both ★ derivations today were the same trick: don't touch densities — push events through monotone maps and read off the CDF.)

Worked: the exponential sampler, by hand

Target $\text{Exp}(\lambda)$: $F(x) = 1 - e^{-\lambda x}$. Invert it — set $u = F(x)$, solve for x :

$$u = 1 - e^{-\lambda x} \quad \Rightarrow \quad e^{-\lambda x} = 1 - u \quad \Rightarrow \quad x = -\frac{\ln(1 - u)}{\lambda}$$

Spot-check $\lambda = 1$: $u = 0.5 \rightarrow x = \ln 2 \approx 0.693$ — half the darts land below 0.693, the exponential's median. The mean (1.0) is further right, dragged by the tail — consistent with the balance-point slide. ✓

Since $1 - U$ is **also** $\mathcal{U}(0, 1)$ (flip the interval), $x = -\ln(U)/\lambda$ works too — that symmetry is practice problem 6, and what NumPy actually implements.

Code: five lines, ten thousand samples

```
lam = 1.0
rng = np.random.default_rng(0)

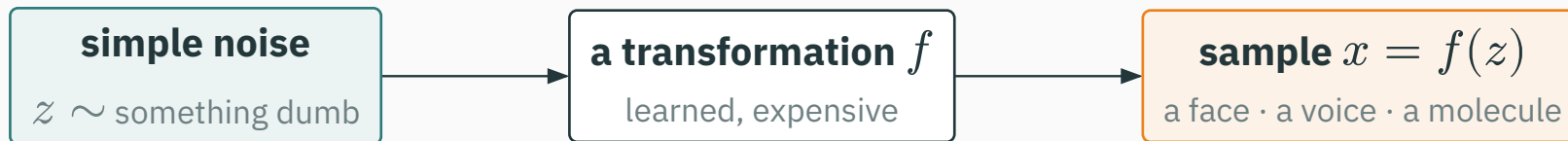
u = rng.random(10_000)          # step 1: the only randomness we own
x = -np.log(1 - u) / lam        # step 2:  $F^{-1}$  — one deterministic line

x.mean()                        # 1.003  $\approx 1/\lambda$   $\checkmark$  (the L12 audit, again)
```

The right panel of the darts figure **is** this code's histogram against $\lambda e^{-\lambda x}$ — sampling theory you can falsify in one plot.

Works verbatim for any distribution whose F^{-1} you can write: uniform, exponential, Laplace, Cauchy, logistic...

The hook, resolved (at cartoon level)



You have now run this pipeline honestly: noise $u \sim \mathcal{U}(0, 1)$, transformation F^{-1} , out came exponential samples — **sample simple noise, transform it** is a theorem, not a slogan.

The industrial versions upgrade each box, not the idea: the noise goes multivariate Gaussian (L13), the transformation becomes **learned** (ES 667), and the toll $|\det J|$ prices every step.

The Gaussian's own F^{-1} has no closed formula — libraries use `erfinv` or the Box–Muller trick. `torch.randn` is still “uniform bits, transformed” under the hood.

Summary & what's next

The discrete → continuous dictionary

	Discrete (ES 114)	Continuous (today)
chance object	PMF: $p(x) = P(X = x)$	pdf: probability per unit length
$P(X = x)$	can be positive	always 0
can exceed 1?	never	happily ($\mathcal{U}(0, 1/2)$ has $p = 2$)
combine	$\sum$	$\int$ — probability is area
CDF	staircase	smooth ramp; $F' = p$
$\mathbb{E}[X]$, Var	$\sum x p(x), \dots$	$\int x p(x) dx, \dots$
transform $Y = g(X)$	relabel outcomes	relabel and pay $ dx/dy $

One table, the whole lecture. The left column you owned already; the right column is now yours too.

Lecture 12 — summary

- **Density** = probability per unit length: $P(X = x) = 0$; $p(x) > 1$ legal; **areas** are probabilities.
- **CDF**: $F(x) = \int_{-\infty}^x p$, and $F' = p$; interval questions are $F(b) - F(a)$.
- **Moments are integrals**: uniform balances at $(a + b)/2$; exponential waits $1/\lambda$.
- **Zoo on sight**: uniform, exponential, Gaussian, Beta (L15's star).
- **The toll**: $p_Y(y) = p_X(x)|dx/dy|$ — the Gaussian's $1/\sigma$ is a receipt.
- **The sampler**: $F^{-1}(U) \sim F$, proved in two lines — uniform darts become anything.

Read before L13 — MML §6.1–6.4 (densities, CDFs, moments) and **§6.7** (today's change of variables, in print). Play: Distribution Explorer · Seeing Theory. **Tutorial this week**: densities and samplers, by hand and in NumPy.

Practice problems

Try on paper; verify against the tutorial notebook.

1. $X \sim \mathcal{U}(2, 6)$: the pdf's height, $P(3 \leq X \leq 4)$, $\mathbb{E}[X]$, $\text{Var}(X)$.
2. The density $p(x) = 3x^2$ on $[0, 1]$: check the area, find F , compute $\mathbb{E}[X]$, and build the sampler $F^{-1}(u)$.
3. Show $\text{Var}(\text{Exp}(\lambda)) = 1/\lambda^2$ by computing $\mathbb{E}[X^2]$ (integrate by parts twice).
4. $Y = -2X + 3$ with $X \sim \mathcal{U}(0, 1)$: find Y 's support and density — mind the $|a|$.
5. Heights in inches $\rightarrow$ centimeters ($\times 2.54$): what happens to the density's height, and to its units? Reconcile with “area = 1”.
6. Why does $-\ln(U)/\lambda$ sample $\text{Exp}(\lambda)$ just as well as $-\ln(1 - U)/\lambda$? Which property of $\mathcal{U}(0, 1)$ did you use?



When F^{-1} is out of reach

★ OPTIONAL

Many densities (most Bayesian posteriors!) have no invertible CDF. Two escape hatches, in cartoon:

- **Rejection sampling** — rain uniform points under an easy envelope curve that covers p ; keep the points landing under p , discard the rest. Kept points are exact samples from p . Cost: the gap between envelope and target is pure waste.
- **Importance sampling** — sample from a wrong-but-easy q , then repair the bias by weighting each sample by $p(x)/q(x)$. Nothing discarded, but weights can degenerate.

Pointers only, by design: MacKay Ch. 29 is the beautiful reference; the probabilistic-ML course (and MCMC) picks up this thread properly. For this course, inverse-CDF + the toll rule are the samplers you own.

Continuous probability is area, and transformations charge a Jacobian toll.

Next: **Multivariate Gaussians** — one bump, many dimensions: the ellipse that runs half of ML.